

BRAC University
Dept. of Computer Science and Engineering
Spring 2025
CSE331-01
QUIZ - 1(A)

Time: 20 minutes

Marks: 20

Name:

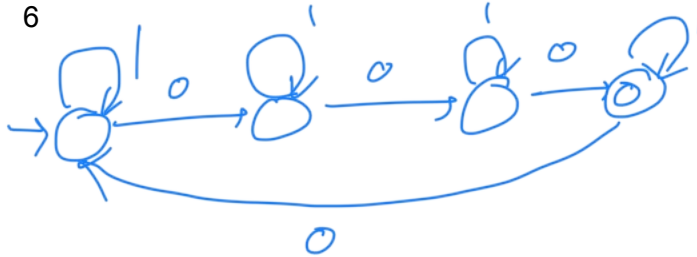
ID:

Section:

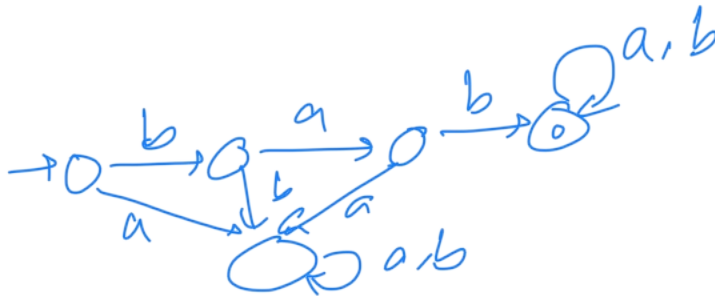
1. Construct DFAs that recognize the following languages.

- a. $L = \{ w \in \{0,1\}^* : \text{the count of 0 in } w \text{ is one less than the multiple of 4} \}$ 7
- b. $L = \{ w \in \{a,b\}^* : w \text{ starts with "bab"} \}$ 7
- c. $L = \{ w \in \{a,b\}^* : w \text{ ends with } n \text{ number of a's, where } n \text{ is a multiple of three} \}$ 6

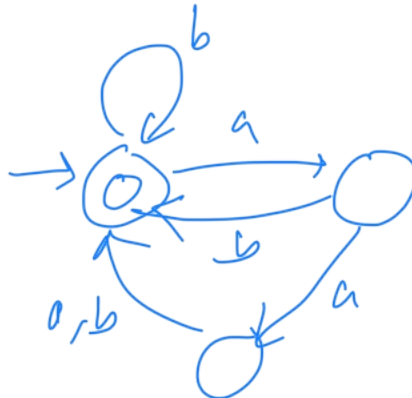
a.



b.



c.



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QUIZ - 1(B)

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2. Construct DFAs that recognize the following languages.

- a. $L = \{ w \in \{0,1\}^* : \text{the count of 1 in } w \text{ is one less than the multiple of 4} \}$ 7
- b. $L = \{ w \in \{a,b\}^* : w \text{ starts with "aba"} \}$ 7
- c. $L = \{ w \in \{a,b\}^* : w \text{ ends with } n \text{ number of b's, where } n \text{ is a multiple of three} \}$ 6